Dr Huy Nguyen

Research Fellow, QUT • PhD (Computer Vision) • Aerial-Ground Person Re-ID

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Research Interests

Deep learning for computer vision; person re-identification and cross-view matching; aerial-ground image/video analysis; cross-modality (visible–infrared) learning; explainable AI; 3D and texture-based representations; benchmark dataset development.

Education

PhD, Electrical Engineering & Computer Science 2021–2026

Queensland University of Technology (SAIVT Lab)

Thesis: Bridging Aerial and Ground Perspectives: Deep Learning for Cross-View Person Re-Identification

Status: Completed — PhD awarded June 2026 (passed with an outstanding examiners' recommendation)

Supervisors: Dr. K. Nguyen, Prof. C. Fookes, Prof. S. Sridharan

BE (Honours), First Class 2017–2021

Queensland University of Technology — Computer & Software Systems (Washington Accord accredited); GPA 6.18/7.0

Selected Publications

8 papers (6 first-author), 150+ citations, h-index 5 — Google Scholar

[1] **H. Nguyen** et al., "AG-VPReID.VIR: Aerial–Ground Video-based Visible-Infrared Person Re-ID," *IEEE IJCB*, 2025. **Best Paper Award.**

[2] **H. Nguyen** et al., "AG-VPReID: A Large-Scale Benchmark for Aerial-Ground Video-based Person Re-ID," *CVPR*, 2025. (A*)

[3] **H. Nguyen** et al., "Aerial-Ground Video-based Person Re-ID with Neural ODE-Enhanced Fusion," *IEEE T-BIOM*, 2026.

[4] **H. Nguyen** et al., "Beyond Geometry: The Power of Texture in Interpretable 3D Person Re-ID," *CVIU*, 2025.

[5] **H. Nguyen** et al., "AG-ReID.v2: Bridging Aerial and Ground Views for Person Re-ID," *IEEE TIFS*, 19:2896–2908, 2024.

Research Experience

Research Fellow — QUT, Electrical Eng. & Robotics May 2026–Present

- Full-time academic appointment (Level A) on an externally funded research project, supervised by Prof. C. Fookes.

PhD Researcher — QUT, SAIVT Lab Feb 2022–Jun 2026

- Built first-of-kind aerial-ground re-ID benchmarks: AG-ReID.v1/v2 (100K+ images), AG-VPReID (9.6M+ frames, 6,632 IDs), AG-VPReID.VIR (first aerial-ground visible-infrared dataset).
- Designed novel architectures: elevated-view attention, memory-based cross-view adaptation, and Neural ODE-guided RGB–infrared fusion.
- Developed a 3D UVTexture framework for viewpoint-invariant, explainable re-ID.
- Released open-source benchmark datasets adopted by the research community.

HDR Research Intern — QUT (funded internship) May–Nov 2024

- Competitive 6-month funded research internship (supervisor Prof. C. Fookes); built the AG-VPReID benchmark and AG-VPReID-Net (three-stream framework) — published at CVPR 2025.

Professional Service

- **Program Co-Chair**, AERO-HPR Workshop, IEEE/CVF CVPR 2026.
- **Guest Editor**, IEEE T-BIOM Special Issue (AERO-HPR; accepted 2026).
- **Co-Organiser & Program Chair**, IJCB Aerial-Ground Re-ID Challenge series (AG-ReID 2023, AG-VPReID 2025).
- **Journal Reviewer**, IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI); Image and Vision Computing (IMAVIS), 2026.

Links

LinkedIn: in/thanh-nhat-huy-nguyen
GitHub: huynguyen792
Scholar: Google Scholar profile
ORCID: 0000-0002-6904-1500

Key Awards

- IEEE IJCB **Best Paper Award**, Osaka (2025)
- QUT Outstanding Doctoral Thesis Award (ODTA) — Nominated (2026)
- QUT HDR Excellence Award (2025)
- Faculty of Engineering High Achiever Award (2025)
- QUT Postgraduate Research Award — QUTPRA (2022–25)
- Golden Key International Honour Society (2017)

Technical Skills

Deep Learning: PyTorch, TensorFlow, CNNs, Transformers, GANs, self-supervised learning, neural ODEs

Computer Vision: person re-ID, object detection (YOLO, Faster R-CNN), segmentation, OpenCV, 3D person representations, video analysis

Programming: Python (expert), C/C++, MATLAB

Tools: Git, Docker, Linux, CUDA, LaTeX

Grants & Funding

- QUTPRA full PhD scholarship, $28,854/yr (2022–25)
- QUT Faculty Write-Up Scholarship (2026)
- ARC Discovery Grant DP200101942, ack. (2020–23)

Languages

Vietnamese (Native)
English (Fluent — IELTS 7.0, C1)
Japanese (Conversational — N4)

Referees

Available on request; supervisors Dr. K. Nguyen, Prof. C. Fookes, Prof. S. Sridharan (QUT SAIVT).